Subject : SECP3213 BUSINESS INTELLIGENCE

Semester : 2/2526

Section : 02

Task : Business Intelligence Group Project

Phase : Visualization and Dashboard (Stage 3)

Dataset Link : https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce/data

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1.0 Introduction

In this phase, the emphasis is on creating an interactive dashboard with Microsoft Power BI. The Olist e-commerce data was cleaned and imported into Power BI for analysis of sales performance, customer distribution, payment behavior, order status, and delivery efficiency.

The Power BI report comprises two dashboards and one storyboard. The sales and customer overview is the focus of dashboard 1. Dashboard 2 is dedicated to payment and delivery performance. The storyboard summarizes the key insights and business recommendations.

Interactive slicers were introduced to enable the user to filter the dashboard by customer state, order status, payment type, payment category, delivery status, and order month.

2.0 Dashboard 1: Sales and Customer Overview

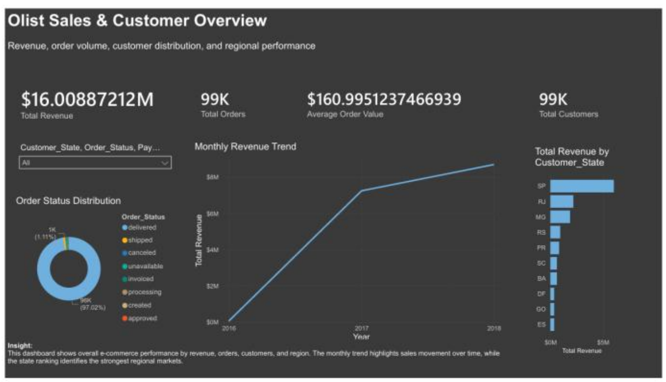

Figure 1: Sales and Customer Overview Dashboard

Dashboard 1 offers a general overview of Olist's business performance. It contains KPI cards for total revenue, total orders, average order value, and total customers. These KPIs help users quickly understand the overall business performance. A trend chart of monthly revenue displays the changes in revenue over time. This can help users determine sales growth, the best months of the year, and the worst months. A line chart was selected as it is appropriate for time-based analysis.

The top states by revenue chart compares regional performance. This will help users determine which states have the most revenue-generating customers. The order status chart displays the status distribution of delivered, cancelled, shipped, unavailable, and other orders. This dashboard can be used to monitor sales and analyse performance by region.

3.0 Dashboard 2: Payment and Delivery Performance

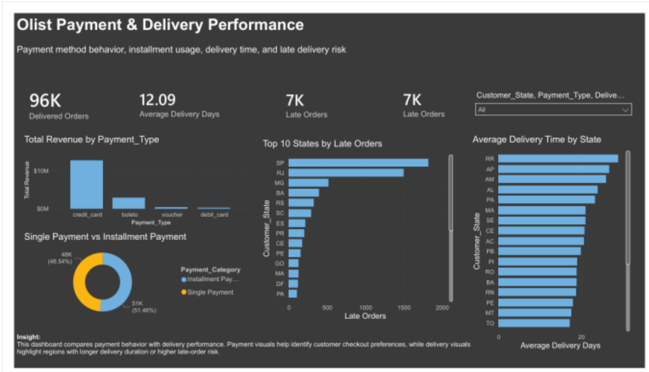

Figure 2: Payment and Delivery Performance Dashboard

Payment behaviour and delivery performance are the focus of Dashboard 2. It contains KPI cards for delivered orders, average days to delivery, late orders, and late order percentage. The revenue by payment type chart displays which payment type is the most profitable. This allows Olist to understand the customers' purchase process. The payment category chart compares single payment and installment payment behavior.

The dashboard provides average delivery time by state and late orders by state for delivery analysis. These charts can be used to identify states that have slower delivery or a higher risk of delivery delay. Only orders that have been delivered and have valid delivery dates are used for delivery visuals to ensure correct analysis.

This dashboard is helpful for improving payment strategy and logistics.This dashboard contains slicers to filter by customer state, payment type, payment category, and delivery status. These filters can be used for deeper analysis, for example, to see whether there are any differences in the late delivery rate between states or differences in revenue by payment type.

The primary purpose of this dashboard is to aid operational decision-making. It assists Olist in understanding payment habits and logistics areas that may need enhancement.

4.0 Business Storyboard

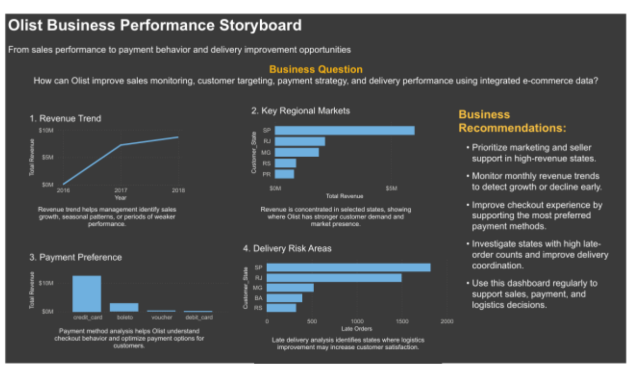

Figure 3: Olist Business Performance Storyboard

The storyboard provides a summary of the key insights drawn from the dashboards. It clearly communicates business questions, insights, and recommendations as a story. The storyboard highlights four main areas: revenue trend, regional performance, payment preference, and delivery risk. It demonstrates how Olist can use the dashboard insights to track sales, understand customer behavior, and optimize delivery operations. The storyboard is helpful because it relates dashboard visuals to business decisions.

5.0 Key Insights

According to the dashboard, Olist has generated approximately **$16.01M** in total revenue from approximately **99K orders** and **99K customers**, with an average order value of approximately **$161**. The delivery of most orders was successful, with approximately **96K** delivered orders, which is about **97.02%** of the total orders. This indicates that Olist has a decent overall order fulfillment rate. Revenue is highly concentrated in the states of São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG). This means that Olist's best customer markets are in a select number of states. Credit cards are the most popular payment method and the highest revenue source for payment behavior. Installment payment is also slightly higher than single payment, with about **51.46%** installment payments compared to **48.54%** single payments. The average delivery time is approximately **12.09 days**. States with the highest late-order counts are mainly **SP, RJ, and MG,** and these states also have high revenue.

6.0 Business Recommendations

It is essential that Olist focus on marketing, seller support, and customer retention in the states that generate the greatest revenue, which are São Paulo, Rio de Janeiro, and Minas Gerais. As credit card is the primary payment method, Olist needs to have a smooth and reliable credit card checkout process. Installment payment should also be well supported, as over half of customers use installment-based payment. Olist should enhance delivery monitoring in SP, RJ, and MG, as these states have the highest number of late orders. These are also high-revenue states, and if there is a delivery delay in these states, it can affect customer satisfaction. Olist should also check states with longer average delivery times, like RR, AP, AM, etc., as they might have logistics issues. In general, Olist needs to make more frequent use of the dashboard for revenue tracking, payment patterns, and delivery problems to make quicker business and logistics decisions.

7.0 Conclusion

Phase 3 was successfully completed by creating an interactive Power BI dashboard for Olist e-commerce analysis. The report includes two dashboards and one storyboard. The dashboards include KPI cards, line charts, bar charts, donut charts, and slicers to facilitate interactive analysis. The dashboard enables Olist to gain insights into sales performance, customer distribution, payment patterns, order status, and delivery efficiency. Thus, it helps improve business intelligence and decision-making.